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### Instruction

What the question is

FSA Construction: 1.6 e, f, h, i, j, l -- build deterministic machines

**1.6** Give state diagrams of DFAs recognizing the following languages. In all parts, the alphabet is  $\{0,1\}$ .

~~g.  $\{w \mid w \text{ contains at least one } 0\}$~~

~~h.  $\{w \mid w \text{ contains at least three } 1\text{s}\}$~~

~~i.  $\{w \mid w \text{ contains the substring } 01 \text{ (i.e., } w = 010 \text{ for some } x \text{ and } y)\}$~~

~~j.  $\{w \mid w \text{ has length at least } 5 \text{ and has at least one } 0\}$~~

**e.**  $\{w \mid w \text{ starts with } 0 \text{ and has odd length, or starts with } 1 \text{ and has even length}\}$

**f.**  $\{w \mid w \text{ doesn't contain the substring } 110\}$

~~k.  $\{w \mid w \text{ has length } 6 \text{ or } 5\}$~~

**h.**  $\{w \mid w \text{ is any string except } 11 \text{ and } 111\}$

**i.**  $\{w \mid \text{every odd position of } w \text{ is a } 1\}$

**j.**  $\{w \mid w \text{ contains at least two } 0\text{s and at most one } 1\}$

~~l.  $\{0, 0\}$~~

**l.**  $\{w \mid w \text{ contains an even number of } 0\text{s, or contains exactly two } 1\text{s}\}$

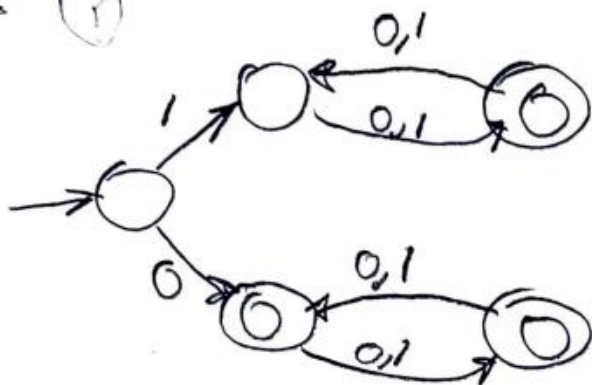
~~m. The empty string~~

~~n. All strings except the empty string~~

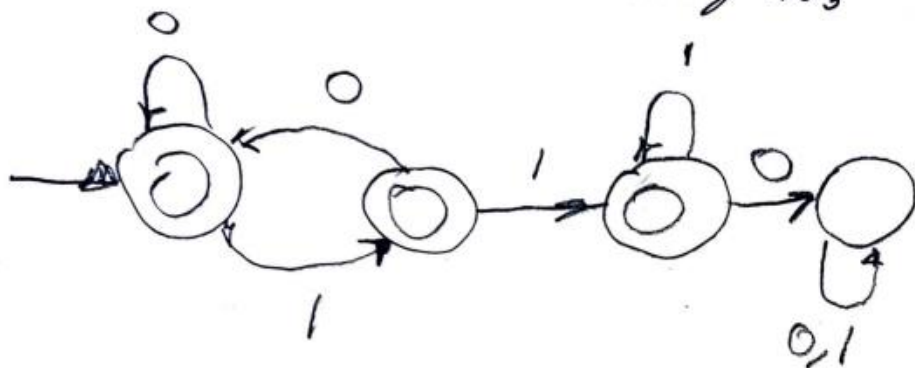
1.6E

(1)

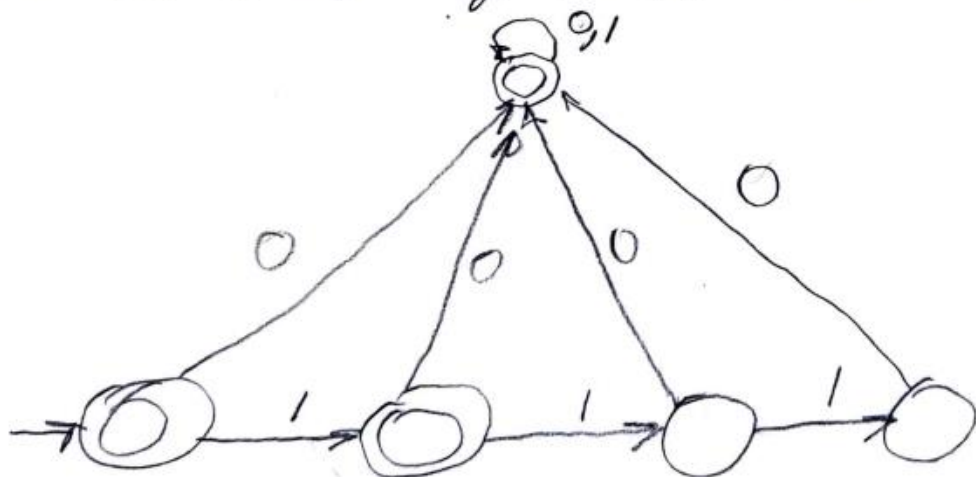
1.6E



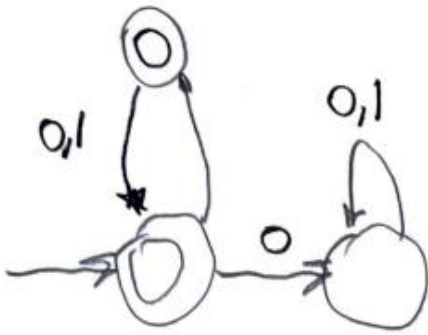
1.6f  $\Sigma^*w\Sigma^*$  doesn't contain the substring 1103



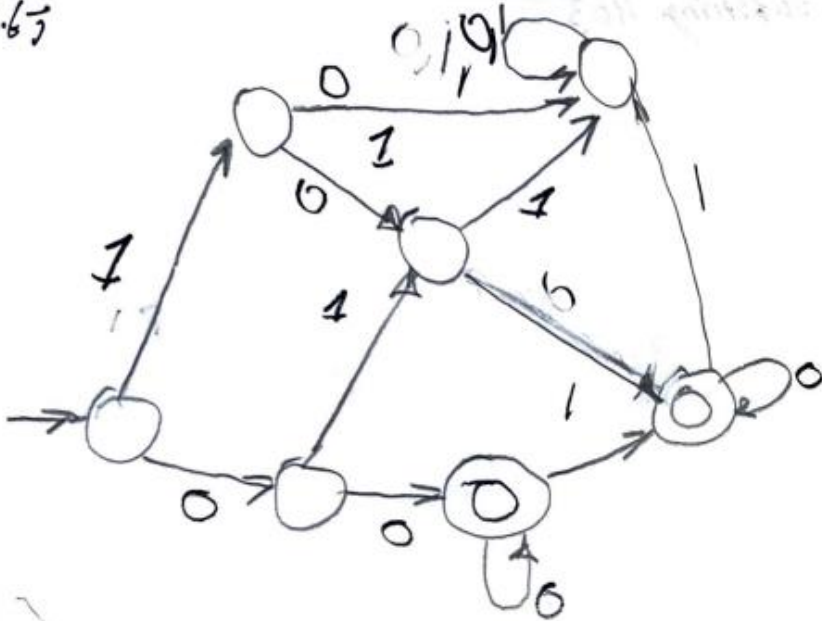
1.6H.  $\Sigma^*w\Sigma^*$  is any string except 11 & 111



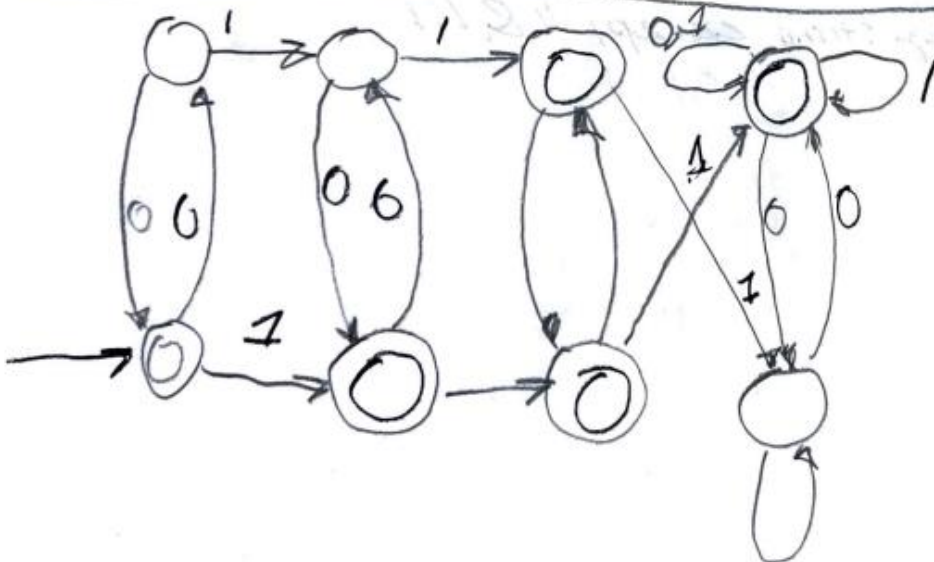
1.65



1.65



1.66



- 1.4 Each of the following languages is the intersection of two simpler languages. In each part, construct DFAs for the simpler languages, then combine them using the construction discussed in footnote 3 (page 46) to give the state diagram of a DFA for the language given. In all parts,  $\Sigma = \{a, b\}$ .

~~$\{w \mid w \text{ has at least three a's and at least two b's}\}$~~

~~A  $\{w \mid w \text{ has exactly two a's and at least one b's}\}$~~

c.  $\{w \mid w \text{ has an even number of a's and one or two b's}\}$

~~A  $\{w \mid w \text{ has an even number of a's and each a is followed by at least one b}\}$~~

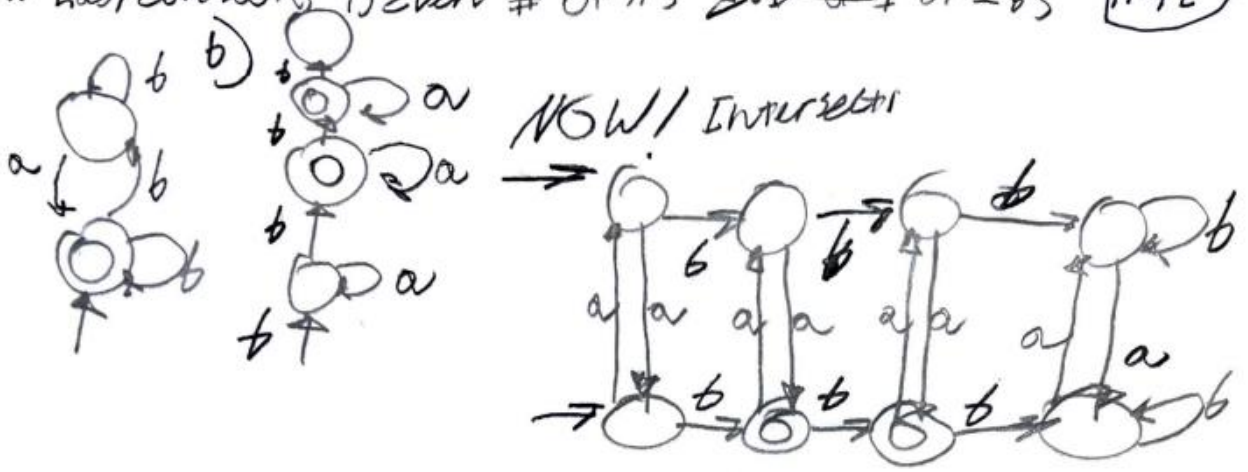
e.  $\{w \mid w \text{ starts with an a and has at most one b}\}$

f.  $\{w \mid w \text{ has an odd number of a's and ends with a b}\}$

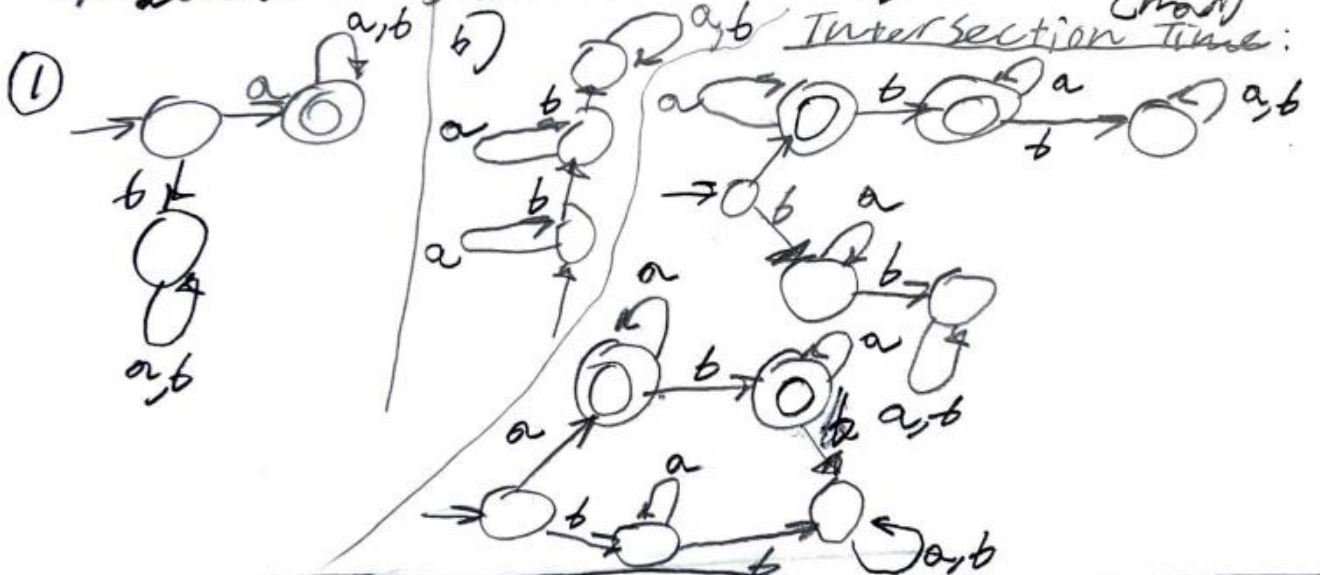
~~g.  $\{w \mid w \text{ has even length and an odd number of a's}\}$~~

w/w has/contains 1) even # of A's 2) ~~ab~~ or 2 b's (1.4C)

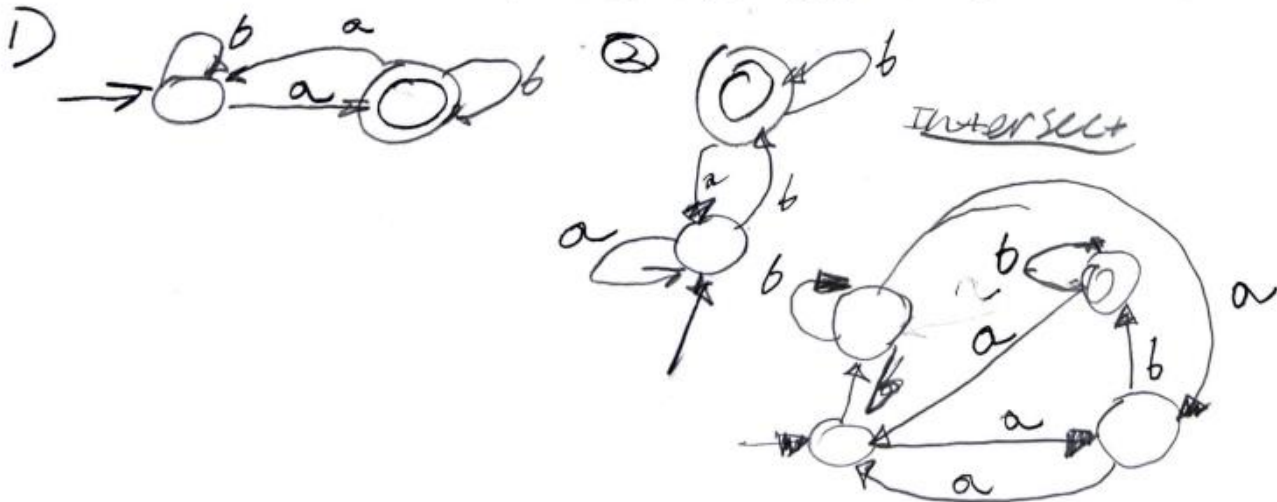
A)



1.4E w/w contains: 1) Starts w/ an A 2) has at most 1 b (max)

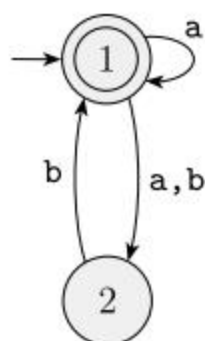


1.4F w/w contains/has 1) odd # of a's 2) end w/ ab

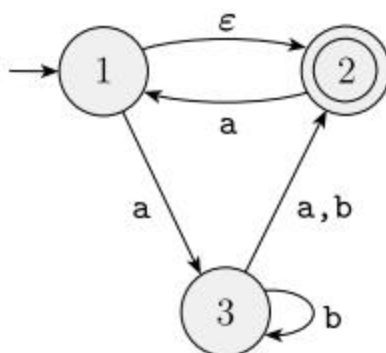




**1.16** Use the construction given in Theorem 1.39 to convert the following two nondeterministic finite automata to equivalent deterministic finite automata.



(a)



(b)

1. Let  $Q' = P(Q)$  where  $Q = \text{subset of all subsets of } Q \rightarrow Q' = \{\emptyset, \{1\}, \{2\}, \{1, 2\}\}$

for Element  $RQ'a$  in set of Alphabet  $\Sigma$ . Cont

$$\delta(R, a) = \{q \in Q \mid q \in \delta(R, a) \text{ for some } r \in R\}$$
$$f(\phi, a) = f(\phi, a) = \phi$$
$$|\langle \Phi, b \rangle = \delta \langle \Phi, b \rangle = 0$$
$$(\{1, 3\}, \alpha) = \delta(1, \alpha) = \{1, 2\}$$
$$(213, 6) = \delta(1, 6) = \{23\}$$
$$N(\{2, 3, a\}) = S(2, a) = \emptyset$$

CS 23, 67 <sup>SL</sup>

$$\{1, 23, a\} = \{1, 23, a\}$$

8 2 4 1, 2 3, 6 →

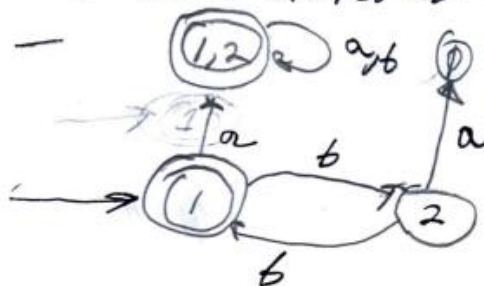
$$3. \varphi_0 = \sum \varphi_0$$

$2C2, \alpha = \emptyset$  kinda obvious it's the same thing written down

$$I_1 = \delta(1, a) \cup \delta(2, a) = \{1, 2\} \cup \emptyset = \{1, 2\}$$

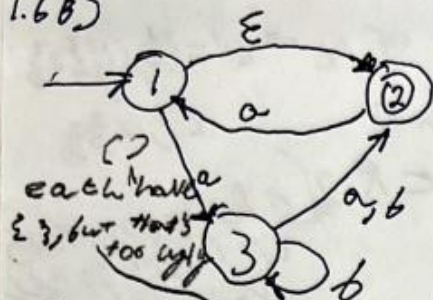
start state NFA  
or accept state NFA

NFA is Present in accept state





1.6B)



Let  $X \subseteq (Q, \Sigma, \delta, q_0, F)$

1.  $Q' = p(Q)$  where  $Q'$  is the subset of all sets of  $Q$   
 $Q' = \{\emptyset, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}, \{1, 2, 3\}\}$   
 $\in$  Notations for each  $R \subseteq Q$   
 $E(R) = \{q \mid q \text{ can be reached from } R \text{ traveling along more } \in \text{ arrow}\}$

$E(\emptyset) = \emptyset$   $E(\{1\}) = 1, 2$   $E(\{2\}) = 2$   $E(\{3\}) = 3$   $E(\{1, 2\}) = 1, 2$   $E(\{1, 3\}) = 1, 2, 3$   
 $E(\{2, 3\}) = 2, 3$   $E(\{1, 2, 3\}) = 1, 2, 3$   
Now do  $\delta(R, a) = \{q \in Q \mid q \in \delta(r, a) \text{ for some } r \in R\}$ . Here  $\delta$  performs the transition on  $r$  for some val of  $a$ .

$\delta(\{1, 3\}, a) = E(\delta(\{1, 3\}, a)) = E(\{3\}) = 3$   
 $\delta(\emptyset, a) = \emptyset$  &  $\delta(\emptyset, b) = \emptyset$

$\delta(\{1, 3\}, b) = E(\delta(\{1, 3\}, b)) = E(\emptyset) = \emptyset$   $\delta(\{2, 3\}, a) = E(\delta(\{2, 3\}, a)) = E(\{2\}) = 2$   
 $\delta(\{2, 3\}, b) = E(\delta(\{2, 3\}, b)) = E(\emptyset) = \emptyset$   $\delta(\{3\}, a) = E(\delta(\{3\}, a)) = E(\{3\}) = 3$   
 $\delta(\{1, 2\}, a) = E(\delta(\{1, 2\}, a)) = E(\{1, 2\}) = 1, 2$

$\delta(\{1, 2, 3\}, a) = E(\delta(\{1, 2, 3\}, a)) = E(\{1, 2, 3\}) = 1, 2, 3$   
 $\delta(\{1, 2, 3\}, b) = E(\delta(\{1, 2, 3\}, b)) = E(\emptyset) = \emptyset$   
 $\delta(\{1, 3\}, a) = E(\delta(\{1, 3\}, a)) = E(\{3\}) = 3$   
 $\delta(\{1, 3\}, b) = E(\delta(\{1, 3\}, b)) = E(\emptyset) = \emptyset$

$\delta(\{2, 3\}, a) = E(\delta(\{2, 3\}, a)) = E(\{2\}) = 2$   
 $\delta(\{2, 3\}, b) = E(\delta(\{2, 3\}, b)) = E(\emptyset) = \emptyset$   
 $\delta(\{1, 2\}, a) = E(\delta(\{1, 2\}, a)) = E(\{1, 2\}) = 1, 2$   
 $\delta(\{1, 2\}, b) = E(\delta(\{1, 2\}, b)) = E(\emptyset) = \emptyset$

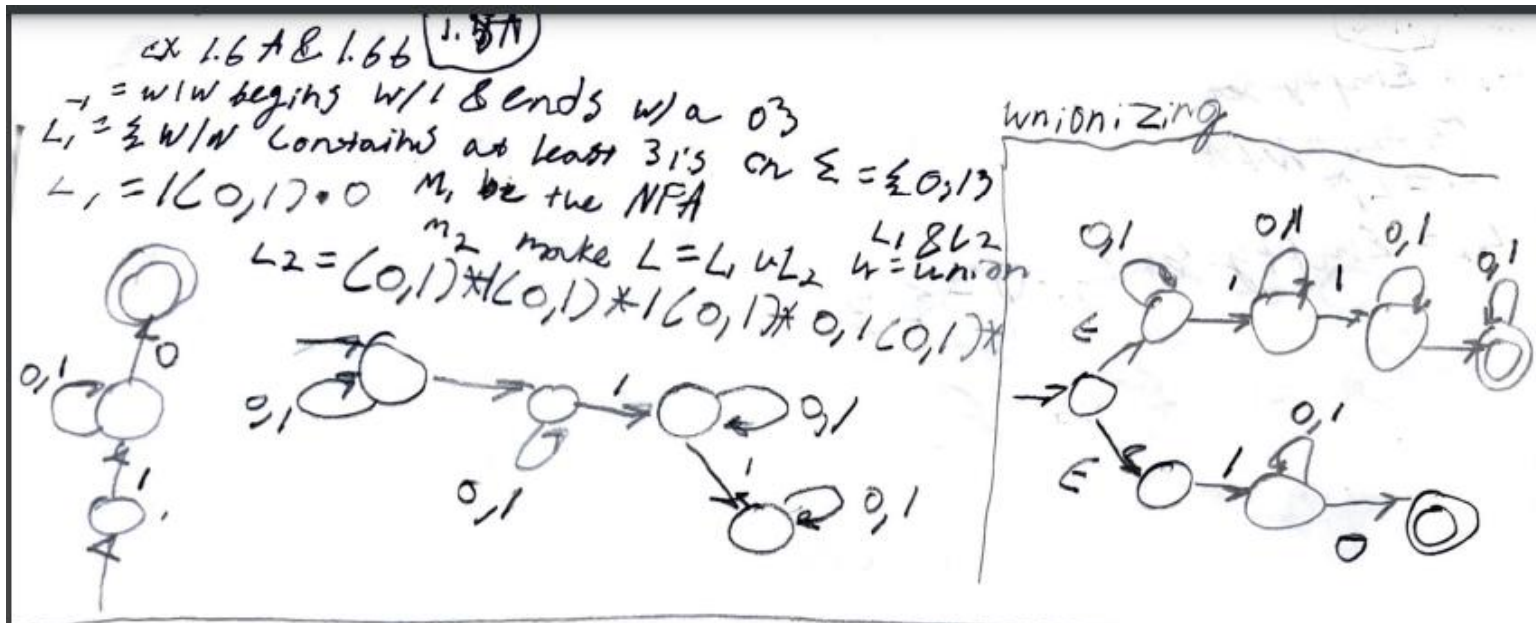
$\delta(\{1, 2, 3\}, a) = E(\delta(\{1, 2, 3\}, a)) = E(\{1, 2, 3\}) = 1, 2, 3$   
 $\delta(\{1, 2, 3\}, b) = E(\delta(\{1, 2, 3\}, b)) = E(\emptyset) = \emptyset$   
 $\delta(\{1, 3\}, a) = E(\delta(\{1, 3\}, a)) = E(\{3\}) = 3$   
 $\delta(\{1, 3\}, b) = E(\delta(\{1, 3\}, b)) = E(\emptyset) = \emptyset$

$\delta(\{2, 3\}, a) = E(\delta(\{2, 3\}, a)) = E(\{2\}) = 2$   
 $\delta(\{2, 3\}, b) = E(\delta(\{2, 3\}, b)) = E(\emptyset) = \emptyset$   
 $\delta(\{1, 2\}, a) = E(\delta(\{1, 2\}, a)) = E(\{1, 2\}) = 1, 2$   
 $\delta(\{1, 2\}, b) = E(\delta(\{1, 2\}, b)) = E(\emptyset) = \emptyset$

$\delta(\{1, 2, 3\}, a) = E(\delta(\{1, 2, 3\}, a)) = E(\{1, 2, 3\}) = 1, 2, 3$   
 $\delta(\{1, 2, 3\}, b) = E(\delta(\{1, 2, 3\}, b)) = E(\emptyset) = \emptyset$   
 $\delta(\{1, 3\}, a) = E(\delta(\{1, 3\}, a)) = E(\{3\}) = 3$   
 $\delta(\{1, 3\}, b) = E(\delta(\{1, 3\}, b)) = E(\emptyset) = \emptyset$



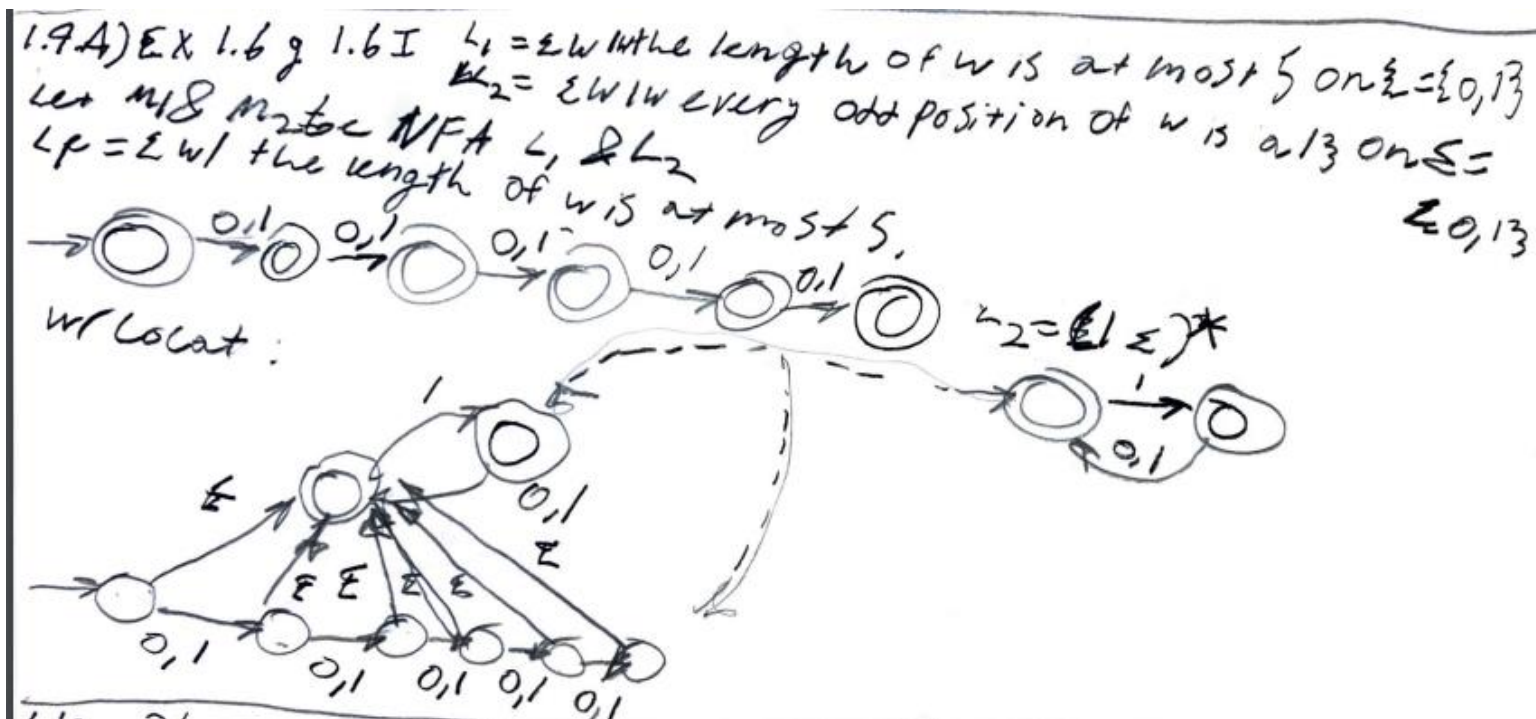




1.9 Use the construction in the proof of Theorem 1.47 to give the state diagrams of NFAs recognizing the concatenation of the languages described in

a. Exercises 1.6g and 1.6i.

~~Exercise 1.6j and 1.6m~~



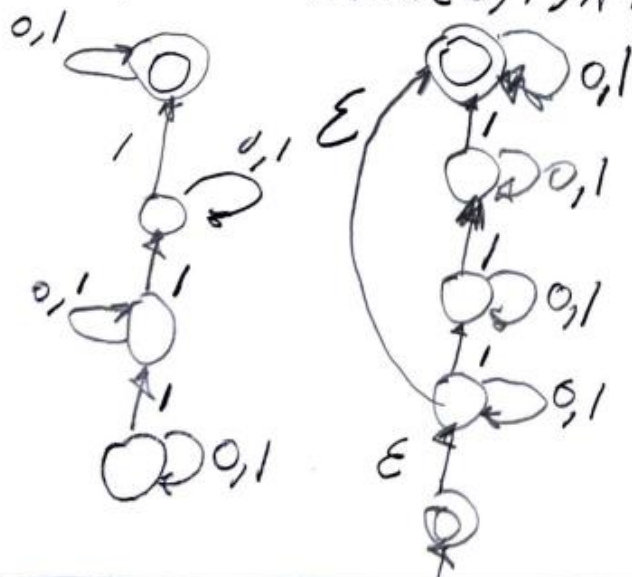
1.10 Use the construction in the proof of Theorem 1.49 to give the state diagrams of NFAs recognizing the star of the languages described in

a. Exercise 1.6b.

~~Exercise 1.6c~~

c. Exercise 1.6m.

1.10 a) Lang  $L_1 = \{w \mid w \text{ contains at least 3 1's}\}$  make  $M_1$  be NFA  $= L_1$ , then  $A$   
 $L = L_1$   
 $L = (0,1)^* 1 (0,1)^* 1 (0,1)^* 1 (0,1)^*$



1.6 · 1.10c)

$L_1 = \text{Empty set}$

$M_1 = L_1 = \text{NFA}$

$L = L_1^*$

$L_1 = \text{Empty set}$   $L_1 = \emptyset = \epsilon$



1.5 Each of the following languages is the complement of a simpler language. In each part, construct a DFA for the simpler language, then use it to give the state diagram of a DFA for the language given. In all parts,  $\Sigma = \{a, b\}$ .

**a.**  $\{w \mid w \text{ does not contain the substring } ab\}$

**b.**  $\{w \mid w \text{ does not contain the substring } baba\}$

**c.**  $\{w \mid w \text{ contains neither the substrings } ab \text{ nor } ba\}$

**d.**  $\{w \mid w \text{ is any string not in } a^*b^*\}$

**e.**  $\{w \mid w \text{ is any string not in } (a^*b^*)^*\}$

**f.**  $\{w \mid w \text{ is any string not in } a^*b^*a^*\}$

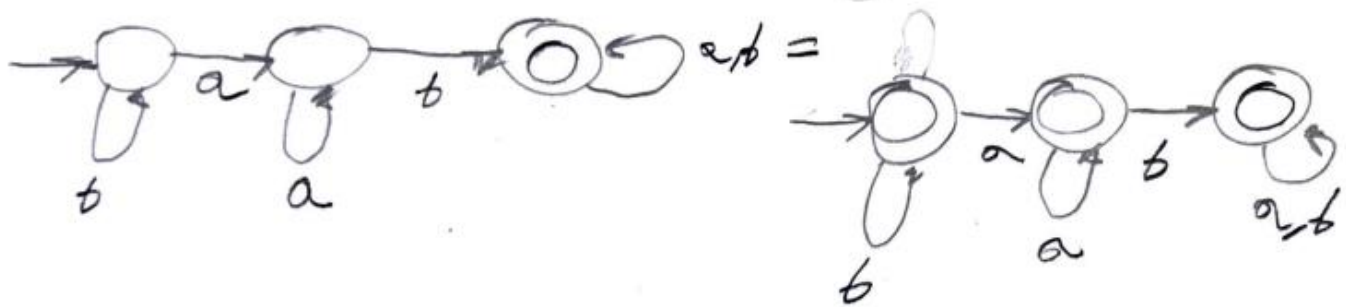
**g.**  $\{w \mid w \text{ is any string that does not contain exactly one } a\}$

**h.**  $\{w \mid w \text{ is any string that does not contain exactly one } b\}$



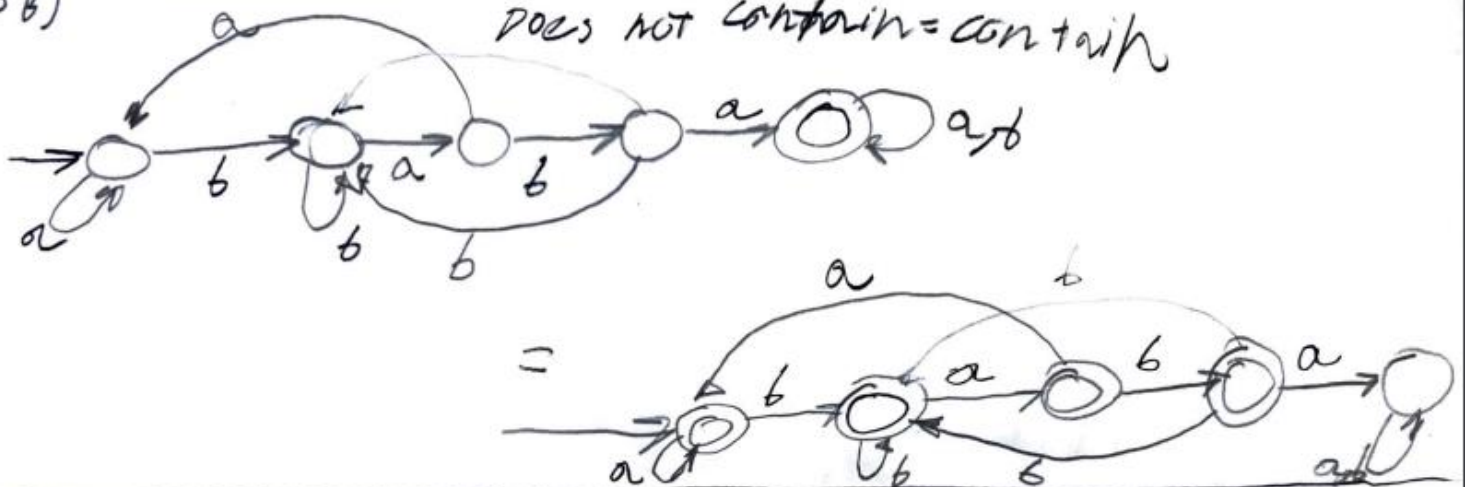
- comp're of simpler Langs  $L$  same for all of them. [1.5A]

Does not contain = contain



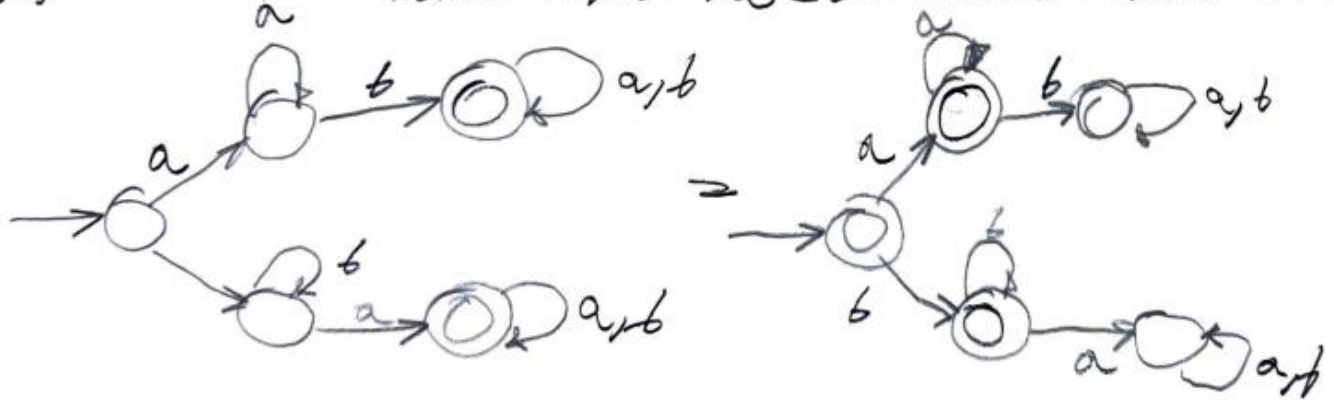
1.5B)

Does not contain = contain



1.5C)

Neither  $a,b$  nor  $b,a$  = contains either  $a,b$  or  $b,a$



1.5D)

any string of  $a^*b^*$

